Cheng Xin

Assistant Professor · California State University, Fresno, Department of Computer Science

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BIO

Cheng Xin is an Assistant Professor of Computer Science at **California State University, Fresno**. He received his Ph.D. in Computer Science from Purdue University under the supervision of Dr. Tamal K. Dey and was a postdoctoral researcher at Rutgers University advised by Dr. Jie Gao. His research develops **trustworthy and theoretically grounded AI systems** by connecting **topology, geometry, algorithms, and multi-agent interaction**. His work spans trustworthy and multi-agent AI, topological and geometric learning, non-Euclidean representations, 3D and video generative AI, and AI for Science, with publications in **NeurIPS, ICML, CVPR, and SoCG**.

RESEARCH INTERESTS

`Topological and Geometric Machine Learning` `Topological Data Analysis`

`3D & Video Generative AI and Evaluation` `Interpretable AI`

`Multi-Agent Safety & Society` `AI for Science`

ACADEMIC APPOINTMENTS & EDUCATION

California State University, Fresno Computer Science — Aug 2026 – Present

Assistant Professor of Computer Science

- Research on **trustworthy and multi-agent AI**, topological and geometric learning, and theoretical foundations for reliable learning systems
- Teaching mathematical foundations of computer science, formal languages, and automata theory

Rutgers University Computer Science — Oct 2023 – May 2026

Postdoctoral Researcher, Advisor: Prof. Jie Gao

- Developed topological frameworks for **interpretable AI**: TopInG for graph learning on molecular benchmarks
- Developed methods for **non-Euclidean representation learning**: Neuc-MDS and Johnson-Lindenstrauss extensions with provable theoretical guarantees
- Contributed to large-scale benchmarks for **3D vision and novel view synthesis**: DL3DV-10K dataset
- Designed algorithms for **multi-agent learning** in social settings

Purdue University Computer Science — Aug 2020 – Aug 2023

Ph.D. in Computer Science, Advisor: Prof. Tamal K. Dey

- Dissertation: Decomposition and Stability of Multiparameter Persistence Modules

The Ohio State University Computer Science & Engineering — Jan 2017 – Aug 2020

Doctoral studies in Computer Science, Advisor: Prof. Tamal K. Dey

- Developed generalized persistence algorithms for multiparameter persistence modules

Lehigh University Computer Science `Jan 2014 – May 2016`

M.S. in Computer Science, Advisor: Prof. Xiaolei Huang

- Thesis: Machine Learning Techniques for Cervigram Image Analysis
- Research Focus: Medical image analysis, machine learning applications

Tongji University Shanghai, China `Sep 2009 – Jul 2013`

B.Eng. in Software Engineering

INVITED TALKS

"TopInG: Topologically Interpretable Graph Learning via Persistent Rationale Filtration"
TDA Conference, University of Missouri · Nov 2025

"Understanding through Shape of Data: Topological Data Analysis for Interpretable AI"
Management Science & Information Systems Colloquium, Rutgers University · Oct 2024

"Exploring Representations Beyond Euclidean Geometry"
John Hopcroft Center Seminar, Shanghai Jiao Tong University · Jun 2024

"Multiparameter Persistence and Its Applications"
CS Theory Seminar, Rutgers University · Nov 2023

"Generalized persistence algorithm for decomposing multi-parameter persistence modules"
Applied Algebraic Topology Network Seminar · Jul 2020

PEER-REVIEWED PUBLICATIONS

[P1] **SoCG 2026** Soham Mukherjee, Shreyas N. Samaga, **Cheng Xin**, Steve Oudot, Tamal K. Dey. D-GRIL: End-To-End Topological Learning with 2-Parameter Persistence. LIPIcs 367, 79:1-79:17.

[P2] **SoCG 2026** Chengyuan Deng, Jie Gao, Kevin Lu, Feng Luo, **Cheng Xin**. Locality Sensitive Hashing in Hyperbolic Space. LIPIcs 367, 39:1-39:19.

[P3] **NeurIPS 2025** Chengyuan Deng, Jie Gao, Kevin Lu, Feng Luo, **Cheng Xin**[†]. Johnson-Lindenstrauss Lemma Beyond Euclidean Geometry. NeurIPS 2025, vol. 38.

[P4] **ICML 2025** **Cheng Xin**, Fan Xu, Xin Ding, Jie Gao, Jiaxin Ding. TopInG: Topologically Interpretable Graph Learning via Persistent Rationale Filtration. ICML 2025, PMLR 267:68847-68869.

[P5] **NeurIPS 2024** Chengyuan Deng, Jie Gao, Kevin Lu, Feng Luo, Hongbin Sun, **Cheng Xin**[†]. Neuc-MDS: Non-Euclidean Multidimensional Scaling Through Bilinear Forms. NeurIPS 2024, vol. 37.

[P6] **ICML 2024** Shahrzad Haddadan, **Cheng Xin**, Jie Gao. Optimally Improving Cooperative Learning in a Social Setting. ICML 2024, PMLR 235:17148-17188.

[P7] **CVPR 2024** Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, **Cheng Xin**, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yawen Lu, Xuanmao Li, Xingpeng Sun, Rohan Ashok, Aniruddha Mukherjee, Hao Kang, Xiangrui Kong, Gang Hua, Tianyi Zhang, Bedrich Benes, Aniket Bera. DL3DV-10K: A Large-Scale Scene Dataset for Deep Learning-based 3D Vision. CVPR 2024, pp. 22160-22169.

[P8] **TMLR 2024** Simon Zhang, **Cheng Xin**, Tamal K. Dey. Expressive Higher-Order Link Prediction through Hypergraph Symmetry Breaking. TMLR, 2024.

[P9] **TAG-ML 2023** **Cheng Xin**, Soham Mukherjee, Shreyas N. Samaga, Tamal K. Dey. GRIL: A 2-parameter Persistence Based Vectorization for Machine Learning. TAG-ML (ICML Workshop), PMLR 221:313-333, 2023.

[P10] **JACT 2022** Tamal K. Dey, **Cheng Xin**[†]. Generalized persistence algorithm for decomposing multiparameter persistence modules. J. Appl. Comput. Topol. 6:271-322, 2022.

[P11] **SoCG 2018** Tamal K. Dey, **Cheng Xin**[†]. Computing Bottleneck Distance for 2-D Interval Decomposable Modules. SoCG 2018, LIPIcs 99, 32:1-32:15.

[P12] **Pattern Recognit. 2017** Tao Xu, Han Zhang, **Cheng Xin**, Edward Kim, L. Rodney Long, Zhiyun Xue, Sameer Antani, Xiaolei Huang. Multi-feature based benchmark for cervical dysplasia classification evaluation. Pattern Recognit. 63:468-475, 2017.

[P13] **MLMI 2015** Tao Xu[*], **Cheng Xin**[*], L. Rodney Long, Sameer Antani, Zhiyun Xue, Edward Kim, Xiaolei Huang. A New Image Data Set and Benchmark for Cervical Dysplasia Classification Evaluation. MLMI (MICCAI Workshop), Springer LNCS 9352:26-35, 2015.

[†] Authors listed alphabetically. · [*] Co-first author.

SUBMITTED MANUSCRIPTS

ArXiv links identify preprint versions of the same manuscripts.

[S1] **ICLR 2027 submission** **Cheng Xin**, Chenqi Guo. PCBench: Physical Commonsense Benchmark for Agentic Video Evaluation and Generation.
Submitted to ICLR 2027, Sep 2026.

[S2] **ICLR 2027 submission** **Cheng Xin**. The Cost of Eigenspace Restriction in Shared Attention Control.
Submitted to ICLR 2027, Sep 2026.

[S3] **ICLR 2027 submission** **Cheng Xin**. Exact Minimum-Event Explanations under Localized Observation Uncertainty.
Submitted to ICLR 2027, Sep 2026.

[S4] **ITCS 2027 submission** Chengyuan Deng, Jie Gao, Kuijun Liu, Kevin Lu, Feng Luo, Puheng Weng, **Cheng Xin**. Navigable Graphs for Low-Dimensional Hyperbolic Spaces.
Submitted to ITCS 2027, Sep 2026.

[S5] **ITCS 2027 submission** **Cheng Xin**. Quantum Query Algorithms for the Constructive Diagonal Ramsey Theorem.
Submitted to ITCS 2027, Sep 2026. arXiv:2609.05812 *(preprint)*.

[S6] **ITCS 2027 submission** **Cheng Xin**. Quantum Query Complexity of Persistence Statistics in Graph Zigzags.
Submitted to ITCS 2027, Sep 2026. arXiv:2609.05813 *(preprint)*.

[S7] **ITCS 2027 submission** **Cheng Xin**. Exact Kernel Transfer to Clique Complexes and the Hardness of Normalized Persistence.
Submitted to ITCS 2027, Sep 2026. arXiv:2610.00075 *(preprint)*.

OTHER PREPRINT

[R1] **Preprint 2021** Tamal K. Dey, **Cheng Xin**[†]. Rectangular Approximation and Stability of 2-parameter Persistence Modules.
Preprint, 2021. arXiv:2108.07429.

TEACHING EXPERIENCE

Instructor, CSCI 60: Foundations of Computer Science, Fresno State · Fall 2026.

Developed notes, slides, and assessments on sets, logic, and inference.

Instructor, CSCI 119: Formal Languages & Automata, Fresno State · Fall 2026.

Developed DFA lessons, proof-based problem sets, tagged PDFs, and LaTeX templates.

Lecturer, CS 513: Algorithms, Rutgers (graduate; 45 students) · Fall 2025.

TA, Data Structures & Algorithms (undergrad; 200 students) · Spring 2023.

TA, Computational Geometry (graduate; 30 students) · Fall 2020.

PROFESSIONAL SERVICE

Area Chair, 2nd Annual Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS 2026), August 2026.

Reviewer, ICLR 2027 (review service in 2026).

Reviewer, ICML, NeurIPS, SoCG.

INDUSTRIAL EXPERIENCE

Electronic Arts (EA) Machine Learning Scientist Intern — May – Aug 2018

Big Data Group · Redwood City, CA

- Large-scale machine learning on **Spark**
- Graph learning on relational database, attributes evaluation and selection, dataset compression

Amazon Software Development Engineer Intern — May – Aug 2015

AWS Infrastructure Group · Seattle, WA

- Data management system for network messages supporting receiving, parsing, storing, and retrieving

REFERENCES

Reference contact information; confidential letters are held in Interfolio Dossier.

Dr. Tamal K. Dey
Professor, Computer Science,
Purdue University
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Dr. Jie Gao
Professor, Computer Science,
Rutgers University
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Dr. Feng Luo
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